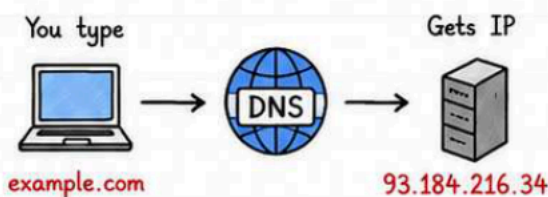




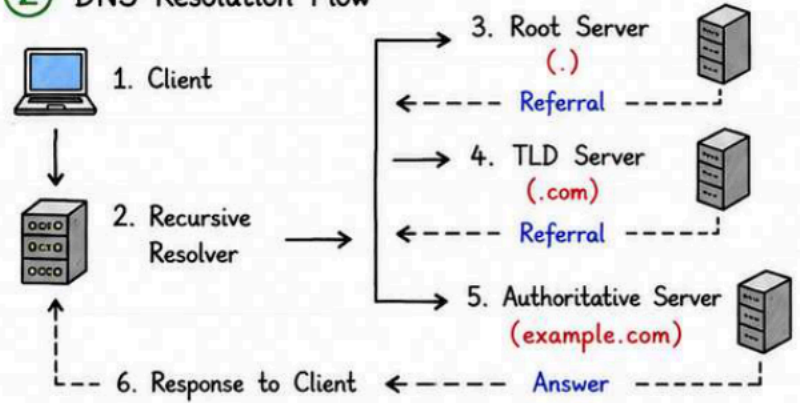
DNS (Domain Name System)

① What is DNS?

DNS is the Internet's phonebook. It translates human-readable domain names into IP addresses that computers use to communicate.



② DNS Resolution Flow



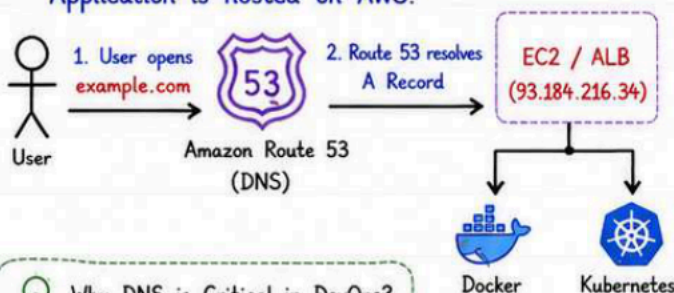
Caching is used to speed up future lookups.

③ Important DNS Record Types

Record Type	Purpose	Example	Explanation
A Record (Address)	Maps a domain name to an IPv4 address.	example.com → 93.184.216.34	Points the domain to an IPv4 address.
CNAME Record (Canonical Name)	Maps one domain name to another domain name.	www.example.com → example.com	Alias of another domain. No IP address.
MX Record (Mail Exchange)	Specifies mail server for the domain.	example.com → mail.example.com (Priority 10)	Tells where to deliver email for the domain.
TXT Record (Text)	Stores text information about the domain.	example.com → "v=spf1 include:_spf.example.com ~all"	Used for SPF, ownership verification, policies, etc.

④ DevOps Example

Application is hosted on AWS.



Why DNS is Critical in DevOps?

- ✓ Maps domains to load balancers / services
- ✓ Enables blue-green / canary deployments
- ✓ Works with CDNs, monitoring, email, etc.

⑤ Quick Reference (Common DNS Record Summary)

Type	Use Case
A	Domain → IPv4 address
AAAA	Domain → IPv6 address
CNAME	Domain alias (one name to another)
MX	Mail server configuration
TXT	SPF, DKIM, verification, policies
NS	Authoritative name servers
PTR	Reverse DNS lookup (IP → Domain)



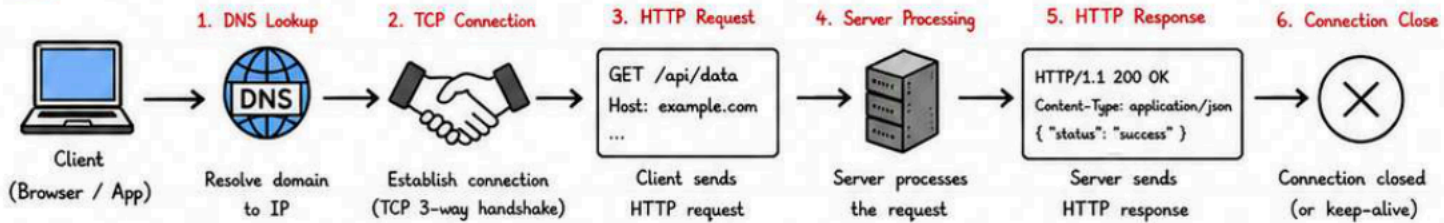
Interview Tip

Always understand how DNS works, record types, and how caching affects resolution time.



HTTP & HTTPS

① Request Lifecycle



Keep-Alive allows multiple requests over the same TCP connection (better performance).

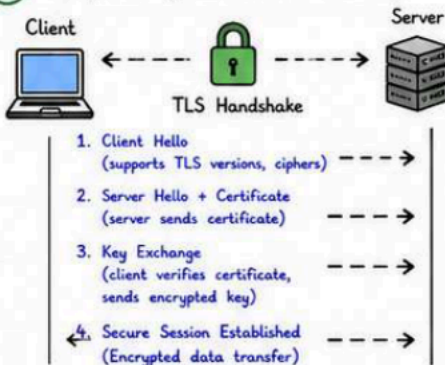
② HTTP Methods

Method	Purpose	Idempotent	Safe
GET	Retrieve data	✓	✓
POST	Create new resource	✗	✗
PUT	Update entire resource	✓	✗
PATCH	Update partial resource	✗	✗
DELETE	Delete resource	✓	✗
HEAD	Get headers only	✓	✓
OPTIONS	Get supported methods	✓	✓

③ HTTP Status Codes (Most Important)

1xx	Informational	Request received, continuing process
2xx	Success	200 OK, 201 Created, 204 No Content
3xx	Redirection	301 Moved Permanently, 302 Found
4xx	Client Error	400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 429 Too Many Requests
5xx	Server Error	500 Internal Server Error, 502 Bad Gateway, 503 Service Unavailable, 504 Gateway Timeout

④ SSL/TLS (How HTTPS Works)



Key Points

- ✓ Uses encryption to protect data in transit.
- ✓ Ensures authentication using SSL Certificate.
- ✓ Ports: HTTP = 80
HTTPS = 443

⑤ REST APIs

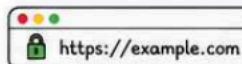
- REST (Representational State Transfer) is an architectural style for APIs.
- Uses HTTP methods to perform operations on resources.
- Resources are identified using URLs.
- Common data format: JSON / XML.

Examples:

GET	/api/users	→ Get all users
POST	/api/users	→ Create user
GET	/api/users/123	→ Get user by ID
PUT	/api/users/123	→ Update user
DELETE	/api/users/123	→ Delete user

⑥ HTTPS Benefits

- ✓ Encrypts data → prevents eavesdropping.
- ✓ Authenticates server → prevents man-in-the-middle attacks.
- ✓ Ensures data integrity → data cannot be tampered.
- ✓ Improves user trust (padlock icon in browser).
- ✓ Required for modern web apps, APIs, payments, logins.



⑦ Quick Reference

Protocol	HTTP (80) / HTTPS (443)
Purpose	Web communication
Data	HTTP → Plain Text HTTPS → Encrypted (TLS)
Security	HTTP → No Security HTTPS → Secure
Best Practice	Always use HTTPS in production



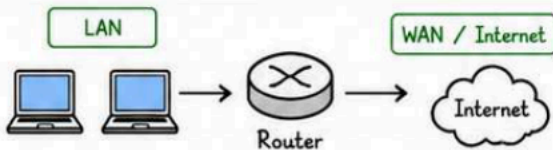
Interview Tip: Explain the request lifecycle, difference between HTTP & HTTPS, and when to use each HTTP method.



Routing, NAT & Gateway

1 Router (What it does)

- A router connects multiple networks and forwards packets based on IP addresses.
- It uses a routing table to decide the best path.



2 Routing (How packets find the path)

Routing Table (Example)

Destination Network	Next Hop
192.168.1.0/24	Directly Connected
10.0.0.0/24	10.0.0.1
0.0.0.0/0	203.0.113.1

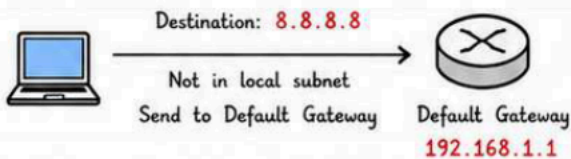


Key Point

0.0.0.0/0 is the Default Route. Used for all unknown destinations (Internet).

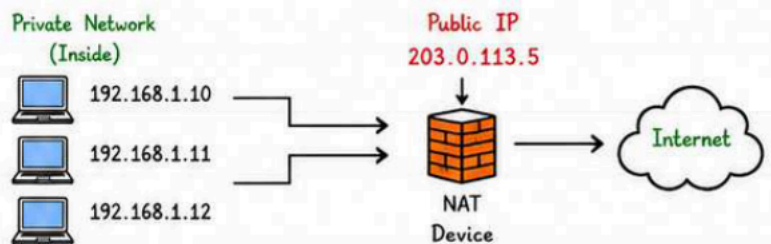
3 Default Gateway

- A default gateway is the router's IP address in your local network.
- If the destination IP is outside the local subnet, traffic is sent to the default gateway.



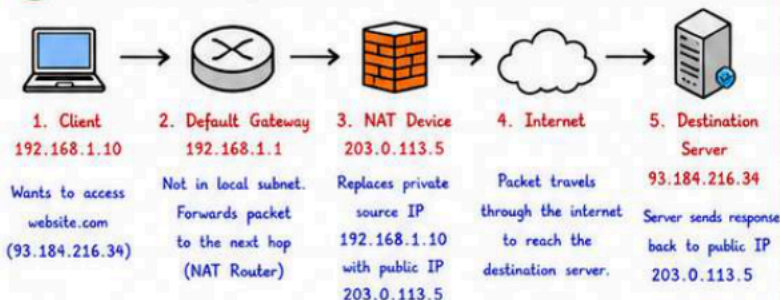
4 NAT (Network Address Translation)

- NAT allows multiple private IPs to share one public IP address.
- Hides internal network from the internet.



✓ **Types:** SNAT (Source NAT) → for outgoing traffic
DNAT (Destination NAT / Port Forwarding) → for incoming traffic

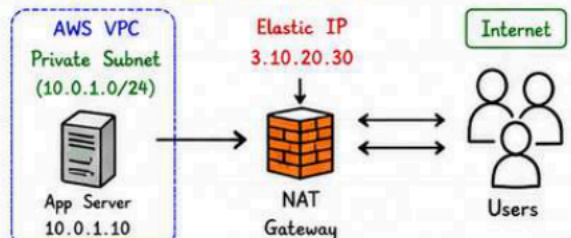
5 Packet Journey (Inside → Internet)



Return Path:
Server → Internet → NAT (203.0.113.5) → 192.168.1.10 (Original Client)

6 Real-World Example

Application is hosted inside AWS VPC and accessed by users over the Internet.



- Users access `https://example.com`
- DNS resolves to 3.10.20.30 (Elastic IP).
- Traffic goes to NAT Gateway → routed to App Server (10.0.1.10).
- Response follows the reverse path.

★ Interview Tip

- ✓ Router decides the path (Routing).
- ✓ Default Gateway = your exit door to other networks.
- ✓ NAT hides private IPs and enables internet access.
- ✓ Understand packet flow end-to-end.

💡 Remember

- ✓ Private IPs: 10.x.x.x, 172.16-31.x.x, 192.168.x.x
- ✓ Default Gateway = Router IP in your subnet
- ✓ NAT = Routing (NAT translates, Router routes)

Common Commands

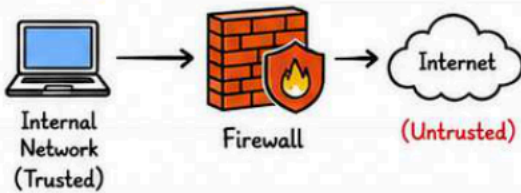
Show Route Table : `ip route (Linux)`
Check Default Gateway : `ip route | grep default`
Test Connectivity : `ping 8.8.8.8`
Trace Route : `traceroute example.com`



Firewalls & Network Security

① What is a Firewall?

A firewall monitors and controls incoming and outgoing network traffic based on security rules.



💡 Firewalls act as a barrier between trusted internal networks and untrusted external networks.

② Stateful vs Stateless Firewall

	Stateless Firewall	Stateful Firewall
	Examines each packet independently.	Tracks the state of active connections.
	Works at L3/L4 (IP, Port, Protocol).	Works at L3/L4 and keeps connection state table.
	Faster but less secure.	Slightly slower but more secure.
	Allows/blocks based only on rules.	Allows return traffic automatically for valid connections.
	Example: Basic ACLs, NACLs.	Example: Most modern firewalls, AWS Security Groups.

③ Security Groups (Stateful)

- Virtual firewall for instances.
- Stateful: Return traffic is automatically allowed.
- Rules are allow only.
- Operates at instance level.
- Example: AWS Security Group, Azure NSG (stateful).

Example (Allow Rules)



Type	Protocol	Port	Source
SSH	TCP	22	203.0.113.10/32
HTTP	TCP	80	0.0.0.0/0
HTTPS	TCP	443	0.0.0.0/0

✓ Only defined traffic is allowed. Everything else is denied.

④ Network ACL (NACL) (Stateless)

- Operates at subnet level (not instance).
- Stateless: You must allow both inbound and outbound.
- Rules are allow and deny.
- Evaluated in order (lowest rule number first).
- Example: AWS NACL, Azure NSG (stateless mode).

Example (Inbound Rules)

Rule #	Type	Protocol	Port	Source	Action
100	HTTP	TCP	80	0.0.0.0/0	ALLOW
110	SSH	TCP	22	203.0.113.10/32	ALLOW
120	ALL	ALL	ALL	0.0.0.0/0	DENY

✓ If a packet doesn't match any ALLOW rule, it's DENIED.

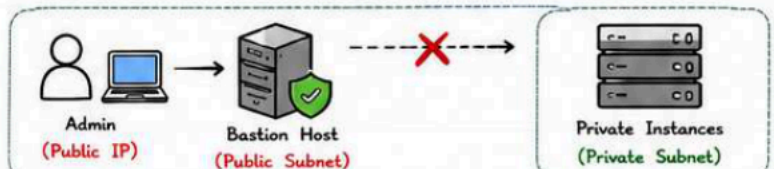
⑤ VPN (Virtual Private Network)

- Creates an encrypted tunnel over the internet.
- Used for secure remote access or site-to-site connectivity.
- Types: Site-to-Site VPN, Client-to-Site VPN.
- Protocols: IPsec, OpenVPN, SSL/TLS.



⑥ Bastion Host (Jump Server)

- A secure gateway host to access private servers.
- Direct access to private instances is blocked.
- Admins SSH/RDP to Bastion first, then to internal servers.
- Best practice: Enable MFA, logging, least privilege access.



★ Key Points

- ✓ Use least privilege - allow only required traffic.
- ✓ Prefer stateful firewalls (SG) for simplicity.
- ✓ NACLs are for subnet-level control & extra layer.
- ✓ Always encrypt traffic using VPN/TLS.
- ✓ Use Bastion Host - never expose private servers.

💡 Interview Tip

- ✓ Stateful firewall remembers connections, stateless does not.
- ✓ Security Groups = stateful, NACL = stateless.
- ✓ Default behavior: SG = Deny all inbound, Allow all outbound.
NACL = Allow all for both (you must deny what you don't want).
- ✓ VPN and Bastion Host improve security posture in cloud.

📄 Common Commands

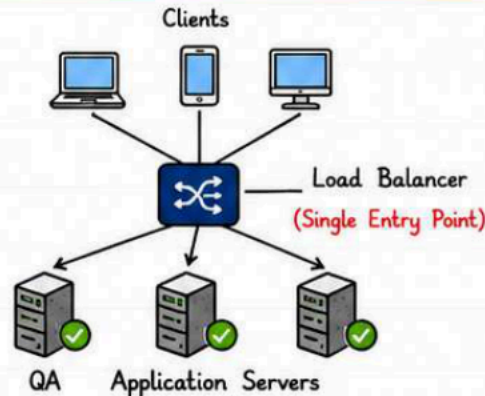
Show Route Table : ip route (Linux)
Check Default Gateway: ip route | grep default
Test Connectivity : ping 8.8.8.8
Trace Route : traceroute example.com



Load Balancer & Proxy

① Why Load Balancer?

- Distributes incoming traffic across multiple servers.
- Improves availability, scalability and performance.
- Prevents single point of failure.
- Performs health checks and routes only to healthy servers.



💡 Key Point

Load balancer hides backend servers, provides high availability and seamless scaling.

② L4 vs L7 Load Balancing

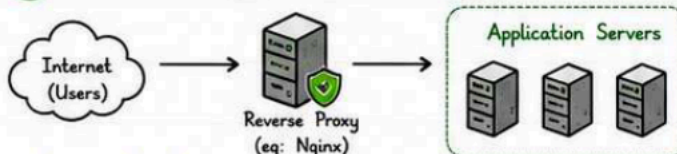
Feature	L4 (Transport Layer)	L7 (Application Layer)
Layer	OSI Layer 4 (TCP/UDP)	OSI Layer 7 (HTTP/HTTPS)
Basis	IP, Port, Protocol	Content (URL, Host, Header)
Speed	Very Fast	Slightly Slower
Example Use	Databases, TCP apps, Microservices	Web apps, APIs, SSL offloading, Content routing
Example Tools	AWS NLB, HAProxy (TCP)	AWS ALB, Nginx, Envoy

③ Forward Proxy (Client-side Proxy)



- Acts on behalf of clients to access the internet.
- Hides internal IPs from external websites.
- Used for caching, filtering, access control.

④ Reverse Proxy (Server-side Proxy)



- Sits in front of backend servers.
- Receives client requests, forwards to appropriate server.
- Provides SSL termination, caching, compression, security.
- Hides backend servers from clients.

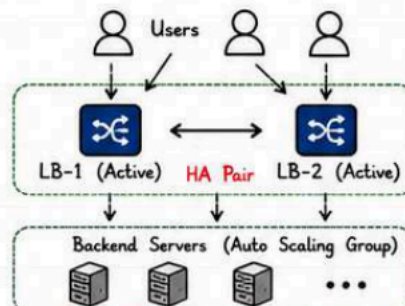
⑤ Nginx (Popular Reverse Proxy / Load Balancer)



- ✓ High performance, lightweight and open-source.
- ✓ Works as Web Server, Reverse Proxy, Load Balancer, API Gateway.
- ✓ Supports L4 & L7 load balancing.
- ✓ SSL termination, caching, compression.
- ✓ Widely used in DevOps, Microservices, Kubernetes Ingress Controllers.

⑥ High Availability (HA) Concepts

- Multiple load balancers (Active-Active)
- Health checks remove unhealthy nodes
- Auto scaling adds/removes servers.
- Use DNS or Anycast IP for LB redundancy.
- Backups, monitoring and alerts are must.



💡 Remember

- ✓ HA = No Single Point of Failure
- ✓ Use multiple components at every Layer (LB, App, DB, Network).
- ✓ Design for failure, not just for performance.

★ Interview Tip

Know the difference between L4 & L7.
Understand when to use Proxy vs Load Balancer.
Be ready with examples: Nginx, HAProxy, AWS ALB/NLB.

🔧 Common Tools

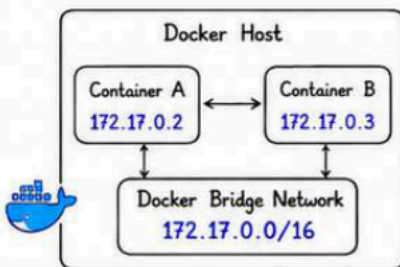
Load Balancers: Nginx, HAProxy, AWS ALB/NLB, GCP Cloud Load Balancer, Azure Load Balancer
Reverse Proxy: Nginx, Traefik, Envoy
Forward Proxy: Squid, Nginx



Docker Networking

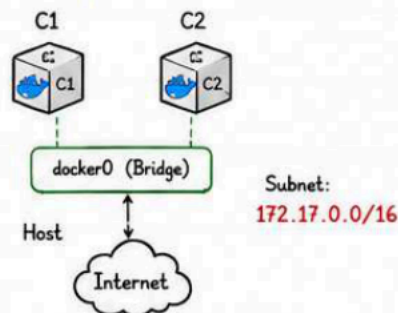
1 Docker Network Overview

- Docker provides multiple network drivers to enable containers to communicate securely and efficiently.
- Use `docker network ls` to list networks.



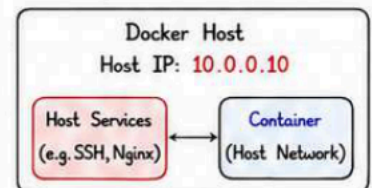
2 Bridge Network (Default)

- Default network mode.
- Containers connect via a virtual bridge (`docker0`) on the host.
- Containers can talk to each other and access external networks through NAT.



3 Host Network

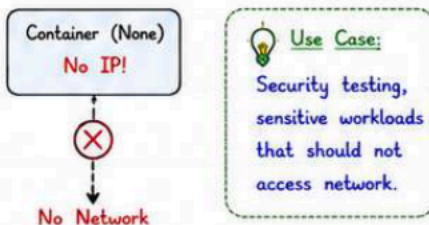
- Container shares the host's network namespace.
- No isolation - container gets host's IP.
- High performance, low latency.



Use Case:
High performance apps, monitoring, network tools.

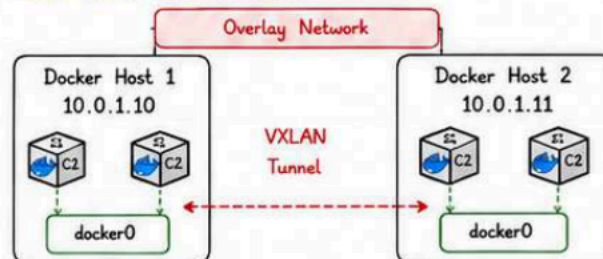
4 None Network

- Container has no networking.
- No IP address, no routing.
- Complete isolation.



5 Overlay Network

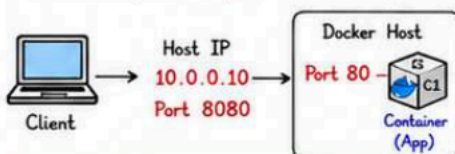
- Used for multi-host container communication.
- Creates a virtual network across Docker hosts.
- Uses VXLAN under the hood.



Use Case:
Swarm mode, Kubernetes, multi-host microservices.

6 Port Mapping (Publish Ports)

- Exposes container ports to the host and external world.
- Format: `-p <host_port>:<container_port>`



Example:
`docker run -d -p 8080:80 nginx`
Access Nginx on `http://10.0.0.10:8080`

7 Docker DNS (Embedded DNS)

- Docker provides an internal DNS server (127.0.0.11) for container name resolution.
- Containers can resolve other containers by name.



Example:
Container A calls: `http://ga-api:8080`
✓ DNS resolves 'ga-api' to 172.17.0.3

★ Key Points

- ✓ Bridge = Default network (isolated but connected)
- ✓ Host = Best performance
- ✓ None = No network access
- ✓ Overlay = Multi-host communication
- ✓ Port Mapping = Expose services
- ✓ Docker DNS = Service discovery by name

★ Interview Tip

Understand when to use each network mode. Know port mapping, container name resolution, and overlay for multi-host setups.

Common Commands

```
docker network ls
docker network inspect <network>
docker run -d --network <network_name> nginx
docker run -d -p 8080:80 nginx
```

Remember

- ✓ Containers on same network communicate by IP or Name.
- ✓ Port mapping exposes container ports.
- ✓ Overlay enables containers on different hosts to talk.



Kubernetes Networking

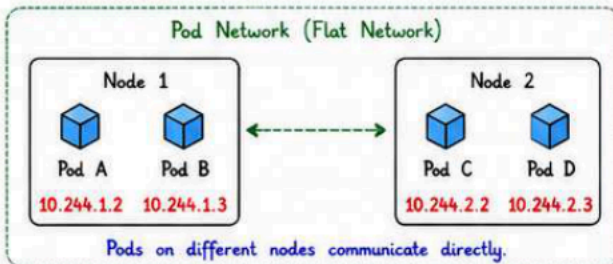
1 Pod Network

- Each Pod gets its own IP address.
- Pods can communicate with each other across nodes without NAT.
- Implemented using CNI plugins (Flannel, Calico, Cilium, Weave).



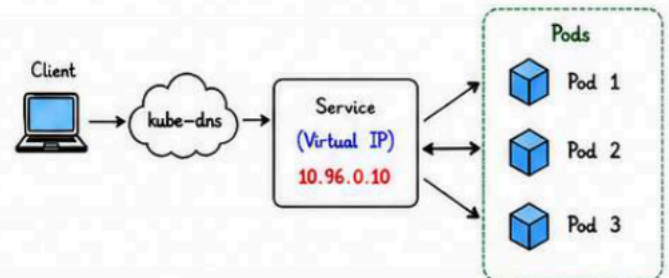
Key Point

Every Pod IP is reachable from every other Pod in the cluster.



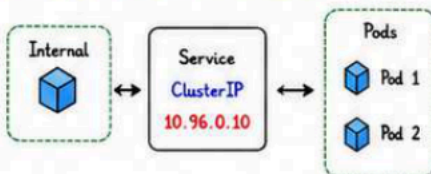
2 Service (Overview)

- Service provides a stable endpoint (IP + DNS) to a set of Pods.
- Decouples clients from Pod lifecycle.
- Kubernetes Service Types:
ClusterIP | NodePort | LoadBalancer | Ingress



3 ClusterIP (Default)

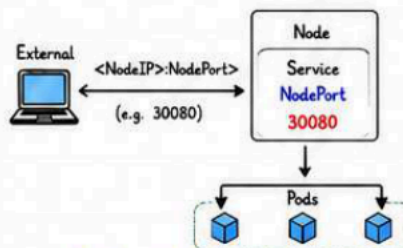
- Exposes the Service on a cluster-internal IP.
- Accessible only inside the cluster.
- Used for internal communication.



Not accessible from outside the cluster.

4 NodePort

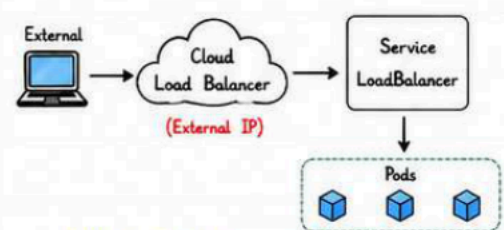
- Exposes the Service on each Node's IP at a static port (30000-32767).
- Accessible from outside via <NodeIP>:<NodePort>.



Traffic goes to Node and then to Pods.

5 LoadBalancer

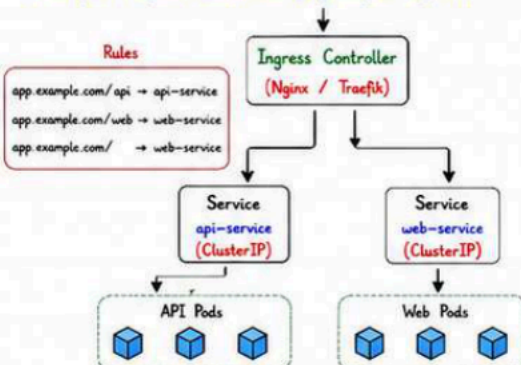
- Creates an external Load Balancer in cloud (AWS ELB, Azure LB, GCP LB).
- Provides a dedicated external IP.



Best for production applications.

6 Ingress

- HTTP/HTTPS routing to Services based on rules.
- Operates at Layer 7 (Application Layer).
- Requires Ingress Controller (e.g., Nginx Ingress).



DevOps Note

Use Network Policies to control Pod-to-Pod traffic for security.
Example: Allow only QA Insights API Pods to access QA Insights DB Pods.

7 Service Types Summary

Type	Exposure	Access	Use Case
ClusterIP	Cluster Internal IP	Inside Cluster	Internal services
NodePort	Node IP + Port	Inside & Outside	Dev / Testing
LoadBalancer	Cloud LB (External IP)	Outside	Production Apps
Ingress	HTTP/HTTPS (L7)	Outside (via rules)	Web Apps / APIs



Interview Tip

- ClusterIP is default and not accessible from outside.
- NodePort exposes on all nodes - be careful with security.
- LoadBalancer is cloud dependent.
- Ingress = smarter routing (L7).
- Pods communicate directly using Pod IPs.



Remember

- Service = Stable endpoint
- Ingress = Smart traffic routing
- Use right Service type based on requirement



Common Commands

```
kubectl get pods -o wide
kubectl get svc
kubectl get ingress
kubectl describe svc <service-name>
kubectl describe ingress <ingress-name>
```



Cloud Networking (AWS VPC)

1 What is VPC?

- A Virtual Private Cloud is a logically isolated network in the cloud.
- You control IP ranges, subnets, route tables, gateways and security.
- Provides secure, scalable and flexible networking for your workloads.

Key Point

VPC is your own private network in the AWS Cloud.

3 Public vs Private Subnets

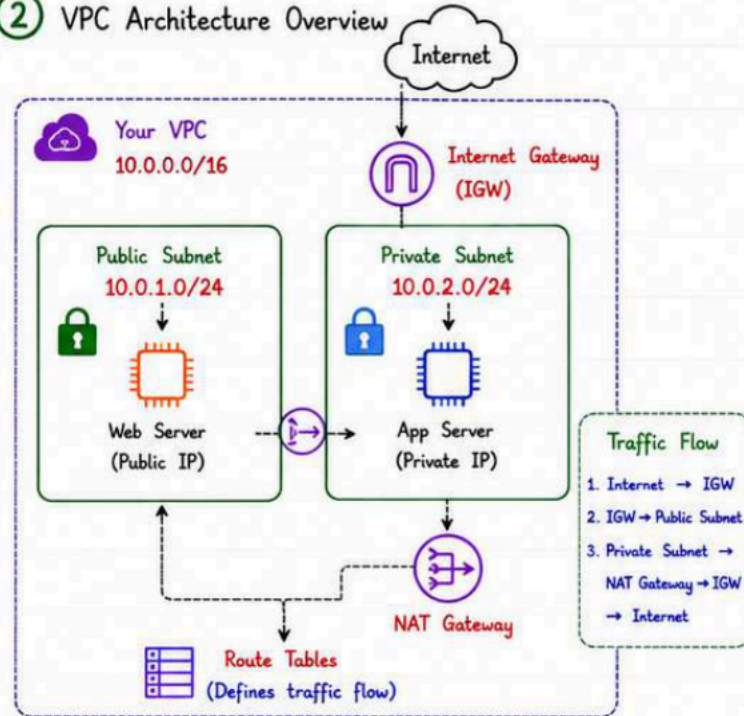
Public Subnet

- ✓ Route to Internet via IGW.
- ✓ Resources have Public IP.
- ✓ Used for bastion, ALB, web servers.

Private Subnet

- ✓ No direct route to Internet.
- ✓ Resources have Private IP.
- ✓ Used for app servers, databases, caching.

2 VPC Architecture Overview



4 Internet Gateway (IGW)

- Horizontally scaled, redundant, and highly available.
- Enables communication between VPC and the Internet.
- Attached to the VPC only.

Example: Internet ↔ IGW ↔ Public Subnet

5 NAT Gateway

- Allows instances in private subnet to access the Internet.
- Blocks inbound connections from Internet.
- Used for updates, downloads, package repositories.

Example: Private Subnet → NAT → IGW → Internet

6 Route Tables

- Defines where network traffic is going.
- Associated with subnets.
- Longest prefix match is used.
- Main Route (local): 10.0.0.0/16 → local

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw-xxxxxx (for Public)
0.0.0.0/0	nat-xxxxxx (for Private)

7 Security Groups (Stateful Firewall)

- Operates at instance level.
- Stateful: return traffic is auto-allowed.
- Allow rules only (implicit deny all).
- Used for fine-grained control.



Example: Allow HTTP(80) and HTTPS(443) to Web Server

8 Best Practices

- ✓ Keep private resources in Private Subnets.
- ✓ Use NAT Gateway for outbound internet access.
- ✓ Use Security Groups for instance-level control.
- ✓ Use separate route tables for different subnets.
- ✓ Enable Flow Logs for monitoring traffic.
- ✓ Use least privilege access.

Interview Tip

- ★ IGW = Internet in
- ★ NAT Gateway = Outbound only for Private
- ★ Route Table decides the path
- ★ Security Group = Stateful, at instance level
- ★ NACL = Stateless, at subnet level
- ★ Public Subnet has route to IGW
- ★ Private Subnet has route to NAT Gateway

Common Use Cases

- ✓ Host public web apps in public subnet.
- ✓ Place application & database in private subnets.
- ✓ Allow private instances to download updates via NAT.
- ✓ Isolate environments (Dev, QA, Prod) in separate VPCs.

Useful Commands (AWS CLI)

- aws ec2 describe-vpcs
- aws ec2 describe-subnets
- aws ec2 describe-route-tables
- aws ec2 describe-security-groups
- aws ec2 describe-nat-gateways
- List VPCs
- List Subnets
- List Route Tables
- List Security Groups
- List NAT Gateways

Remember

- ✓ VPC = Your private network
- ✓ IGW = Internet access
- ✓ NAT = Outbound internet
- ✓ Route Table = Traffic path
- ✓ SG = Stateful firewall



Networking Cheat Sheet

1 Key Comparisons

A. OSI vs TCP/IP

OSI Model (7 Layers)	TCP/IP Model (4 Layers)	Function
7. Application	Application	Network services to applications
6. Presentation		
5. Session		
4. Transport	Transport	End-to-end communication
3. Network	Internet	Logical addressing & routing
2. Data Link	Network Access	Framing, MAC addressing
1. Physical		Bit transmission over media

B. TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Reliable, guaranteed delivery	Best-effort, no guarantee
Order	Maintains order of packets	No order guarantee
Speed	Slower (overhead for reliability)	Faster (low overhead)
Data Type	Stream of bytes	Datagrams (packets)
Use Cases	Web, Email, File transfer, SSH	DNS, VoIP, Gaming, Streaming
Example Ports	80, 443, 22, 3306	53, 123, 161, 500

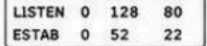
C. HTTP vs HTTPS

Feature	HTTP	HTTPS
Full Form	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Port	80	443
Security	Not Encrypted	Encrypted (SSL/TLS)
Data	Plain Text	Encrypted
Authentication	No	Yes (Certificate)
SEO / Trust	Lower	Higher (Secure)
Use	General web traffic	Sensitive data, APIs, logins



Tip: Use HTTPS everywhere. It encrypts data and builds trust.

2 Common Commands

Command	Purpose	Example	What it shows
ping  	Test reachability and latency.	ping example.com [Windows: ping example.com]	Whether host is reachable, response time (latency) and packet loss.
tracert ① → ② → ③ → ... → ④	Trace the path packets take to reach destination.	tracert example.com [Windows: tracert example.com]	All network hops (routers) and latency at each hop.
nslookup  www.example.com →	Query DNS to get domain information.	nslookup example.com	IP address, Name servers and DNS records.
dig  example.com →	Advanced DNS lookup (tool for admins).	dig example.com dig @8.8.8.8 example.com A	Detailed DNS records, TTL, response codes, etc.
curl  → 	Transfer data to/from server (HTTP/HTTPS).	curl -I https://example.com/api curl -X GET https://example.com/api	HTTP headers, response code and response data.
ip 	Show / configure network interfaces and routes.	ip a ip route	IP addresses, interfaces, routing table.
ss  LISTEN 0 128 80 ESTAB 0 52 22	Show socket statistics (TCP/UDP connections).	ss -tln ss -tp	Listening ports, open connections, process info.

3 Quick Notes



Important Ports (Most Used)

22	SSH (Secure Shell)	5432	PostgreSQL
53	DNS (Domain Name System)	27017	MongoDB
80	HTTP (Web)	6379	Redis
443	HTTPS (Secure Web)	25	SMTP (Email)
3306	MySQL	21	FTP



Remember

- ✓ DNS translates domains to IPs.
- ✓ IP routes packets to destination.
- ✓ TCP = Reliable, UDP = Fast.
- ✓ Firewalls control what is allowed.
- ✓ Always use HTTPS for security.



Best Practices

- ✓ Use least privilege access.
- ✓ Keep security groups tight.
- ✓ Monitor network traffic & logs.
- ✓ Encrypt data in transit (HTTPS/TLS).
- ✓ Document network architecture.